

Regulation (EU) 2024/1689: The Artificial Intelligence Act

Official Legal Framework on Artificial Intelligence & Harmonized Rules across the European Union

Chapter 1: Subject Matter, Scope, and Risk-Based Classification

The Artificial Intelligence Act (Regulation (EU) 2024/1689) establishes a comprehensive, harmonized legal framework for the development, placement on the market, putting into service, and use of artificial intelligence systems in the European Union.

The regulation follows a strict risk-based approach, categorizing AI systems into four distinct risk tiers: Unacceptable Risk (strictly prohibited), High Risk (subject to strict mandatory compliance, risk management, and conformity assessment), Specific Transparency Risk (subject to information and labeling obligations), and Minimal/No Risk (free development and use without additional legal burdens).

The regulation applies to providers placing AI systems or General-Purpose AI (GPAI) models on the EU market, regardless of whether the providers are established within the Union or in a third country, as well as deployers of AI systems who have their place of establishment or are located within the European Union.

Chapter 2: Prohibited Artificial Intelligence Practices (Article 5)

Article 5 explicitly prohibits AI practices deemed unacceptable and contrary to fundamental human rights, public safety, and human dignity. Any deployment of the following AI practices within the EU is strictly illegal:

- **Cognitive Behavioral Manipulation:** AI systems deploying subliminal techniques beyond a person's consciousness, or purposefully manipulative or deceptive techniques that impair a person's ability to make an informed decision, causing significant harm.
- **Vulnerability Exploitation:** AI systems exploiting vulnerabilities of specific persons or groups due to age, disability, or specific socio-economic situation to distort behavior causing significant harm.
- **Social Scoring:** AI systems used by public authorities or on their behalf for evaluating or classifying natural persons based on their social behavior or personality characteristics, leading to detrimental treatment in unrelated social contexts.
- **Individual Predictive Policing:** AI systems making risk assessments of natural persons to assess or predict the likelihood of committing a criminal offense based solely on personality profiling or characteristics.
- **Untargeted Facial Recognition Scraping:** AI systems creating or expanding facial recognition databases through untargeted scraping of facial images from the internet or CCTV footage.
- **Emotion Recognition in Workplace and Education:** AI systems inferring emotions of natural persons in workplace environments and educational institutions, except for strictly medical or safety reasons.
- **Biometric Categorization of Sensitive Attributes:** AI systems categorizing natural persons individually based on biometric data to deduce or infer their race, political opinions, trade union membership, religious or philosophical beliefs, or sexual orientation.
- **Real-Time Remote Biometric Identification:** Real-time remote biometric identification systems in publicly accessible spaces for law enforcement, except under narrowly defined judicial authorizations for imminent threats or serious crimes.

Chapter 3: High-Risk AI Systems Classification (Article 6 & Annex III)

AI systems are classified as High-Risk if they are intended to be used as safety components of products subject to EU harmonized safety legislation (e.g., medical devices, machinery, civil aviation) or if they fall under the standalone critical areas listed in Annex III:

1. Biometric identification and categorization of natural persons (where not prohibited under Article 5).
2. Critical Infrastructure: Safety components in the management and operation of critical digital infrastructure, road traffic, and the supply of water, gas, heating, and electricity.
3. Educational and Vocational Training: Systems used to determine admission, assign students, or evaluate learning outcomes.
4. Employment and Workers Management: Systems used for recruitment, screening applicants, making promotion/termination decisions, and monitoring employee performance.
5. Access to Essential Private and Public Services: Systems evaluating eligibility for public assistance benefits, credit scoring of natural persons, and emergency healthcare dispatch risk prioritization.
6. Law Enforcement: Systems evaluating reliability of evidence, assessing recidivism risks, or profiling individuals during investigations.
7. Migration, Asylum, and Border Control: Polygraphs, verification of travel documents, and examination of asylum applications.
8. Administration of Justice and Democratic Processes: Systems assisting judicial authorities in researching and interpreting facts and the law.

Chapter 4: Mandatory Requirements for High-Risk AI Systems (Articles 9-15)

High-Risk AI systems must comply with rigorous mandatory requirements throughout their entire lifecycle:

- Risk Management System (Article 9): Continuous, iterative risk management process identifying known and foreseeable risks, evaluating risks under intended and foreseeable misuse conditions, and adopting suitable mitigation measures.
- Data and Data Governance (Article 10): Training, validation, and testing datasets must be subject to appropriate data governance practices, addressing bias detection, data collection processes, and data representativeness.
- Technical Documentation (Article 11): Comprehensive documentation drawn up before the system is placed on the market, demonstrating compliance with all legal requirements for national competent authorities.
- Record-Keeping and Automatic Logging (Article 12): Automatic logging capabilities throughout the system's lifetime to ensure traceability of operations, monitoring of performance, and post-market surveillance.
- Transparency and Provision of Information (Article 13): High-risk systems must be designed to enable deployers to understand the system's output, interpret results, and use the system appropriately with clear instructions for use.
- Human Oversight (Article 14): Systems must be designed so that natural persons can oversee operations, prevent or minimize risks, remain aware of automation bias, and override or stop the system at any point (kill-switch capability).
- Accuracy, Robustness, and Cybersecurity (Article 15): Systems must achieve resilient levels of accuracy, robustness against adversarial attacks, input manipulation, data poisoning, and cybersecurity vulnerabilities.

Chapter 5: Transparency Obligations for Certain AI Systems (Article 50)

To safeguard public discourse and prevent deception, Article 50 imposes strict transparency obligations:

- AI-Generated Content Disclosure: Providers of AI systems, including generative AI systems generating audio, image, video, or text content, must ensure outputs are marked in a machine-readable format and detectable as artificially generated or manipulated.

- **Deepfakes Transparency:** Deployers of an AI system that generates or manipulates image, audio, or video content constituting a 'deepfake' must disclose that the content has been artificially generated or manipulated in a visible, clear manner.
- **Emotion Recognition Notification:** Deployers of emotion recognition systems or biometric categorization systems must inform natural persons exposed thereto of the system's operation.

Chapter 6: General-Purpose AI (GPAI) Models (Articles 51-55)

The AI Act introduces specific governance rules for General-Purpose AI (GPAI) foundation models:

- **Baseline GPAI Obligations:** Providers of all GPAI models must maintain technical documentation, provide information to downstream AI providers integrating the model, and respect EU copyright laws.
- **GPAI Models with Systemic Risk:** A GPAI model is classified as having systemic risk if it possesses high-impact capabilities or if the cumulative computational power used for its training exceeds 10^{25} floating-point operations (FLOPs).
- **Systemic Risk Mitigations:** Providers of systemic GPAI models must perform model evaluation, conduct and document continuous adversarial testing (red-teaming), track and report serious incidents to the European AI Office, and maintain state-of-the-art cybersecurity protection.

Chapter 7: Enforcement, Penalties, and Administrative Fines (Article 99)

Article 99 establishes severe, tiered financial penalties for non-compliance with the AI Act:

1. **Prohibited AI Practices Violations (Article 5):** Administrative fines of up to 35,000,000 EUR or, if the offender is an undertaking, up to 7% of its total worldwide annual turnover for the preceding financial year, whichever is higher.
2. **Non-Compliance with High-Risk Obligations (Articles 9-15):** Administrative fines of up to 15,000,000 EUR or up to 3% of total worldwide annual turnover, whichever is higher.
3. **Supplying Misleading or False Information:** Fines of up to 7,500,000 EUR or up to 1.5% of total worldwide annual turnover, whichever is higher.

For SMEs and start-ups, the fines are subject to proportionate caps based on the lower percentage threshold.